Phase 6: User Interface Development

Doctor Appointment & Consultation System (Healthcare CRM)

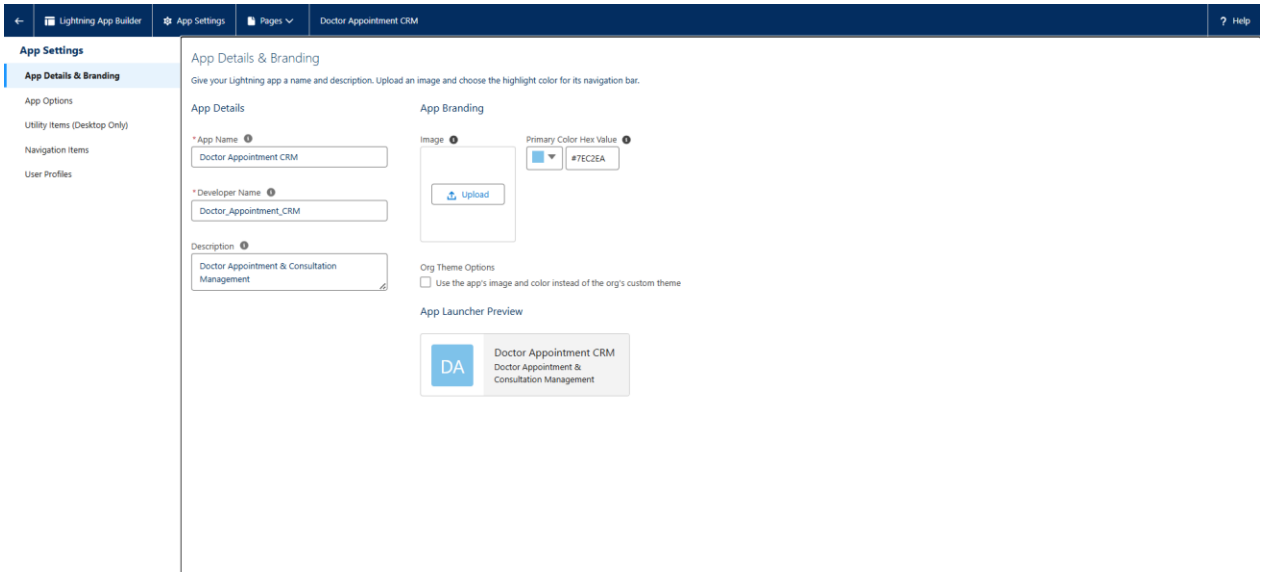
Objective

The goal of this phase is to enhance the user experience of the *Doctor Appointment & Consultation System* by creating customized Lightning pages, apps, and simple LWC components to make the system more interactive and user-friendly for doctors, patients, and administrators.

Components Implemented

1. Lightning App Builder

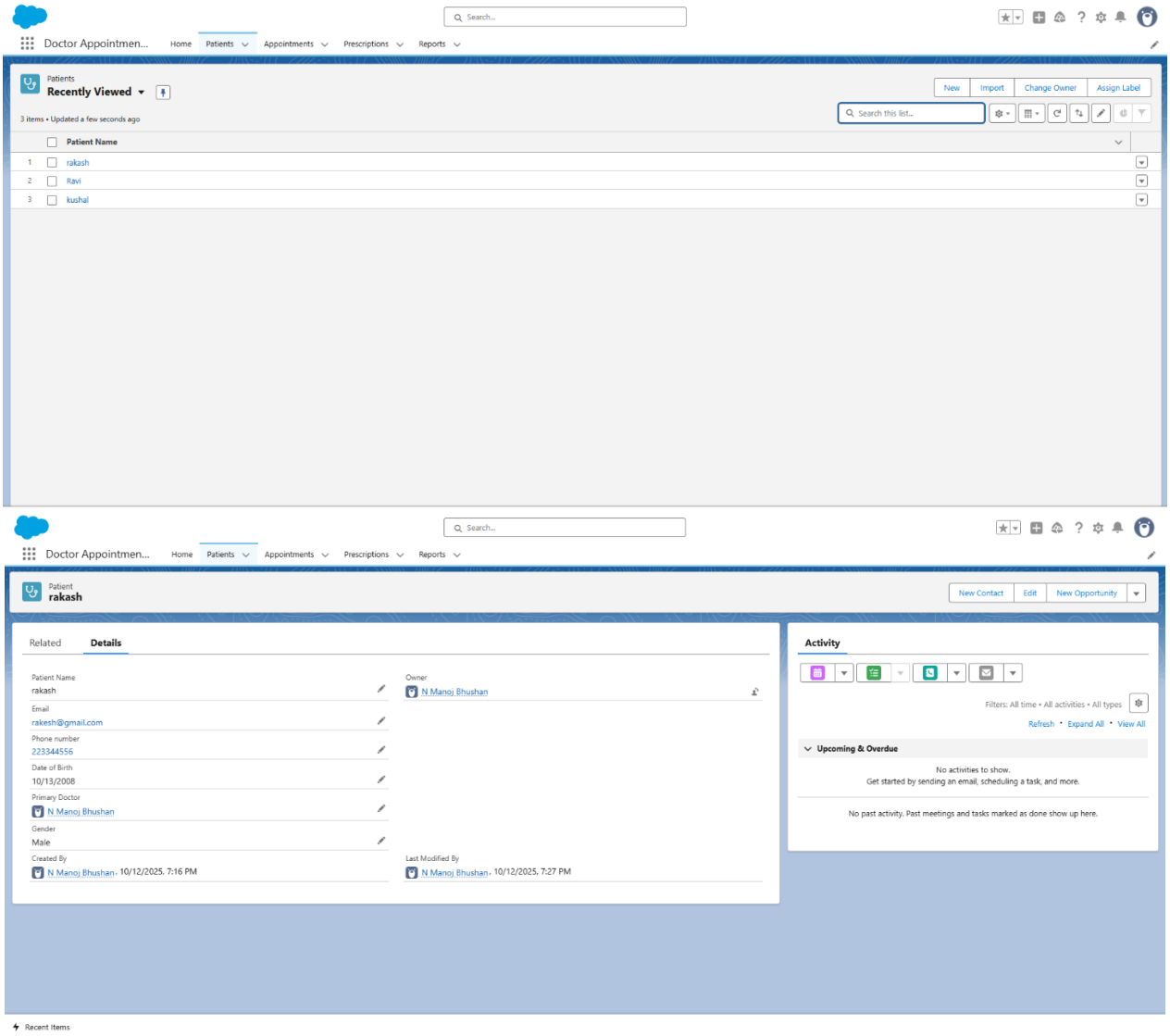
- A custom Lightning App named **"Doctor Appointment CRM"** was created using the **App Manager**.
- The app contains navigation tabs for:
    - **Patients**
    - **Appointments**
    - **Prescriptions**
    - **Reports**
- App branding was customized with the logo and color theme relevant to the healthcare domain.



2. Record Pages

- Custom Record Pages were designed for **Patient** and **Appointment** objects using **Lightning App Builder**.
- Key highlights:

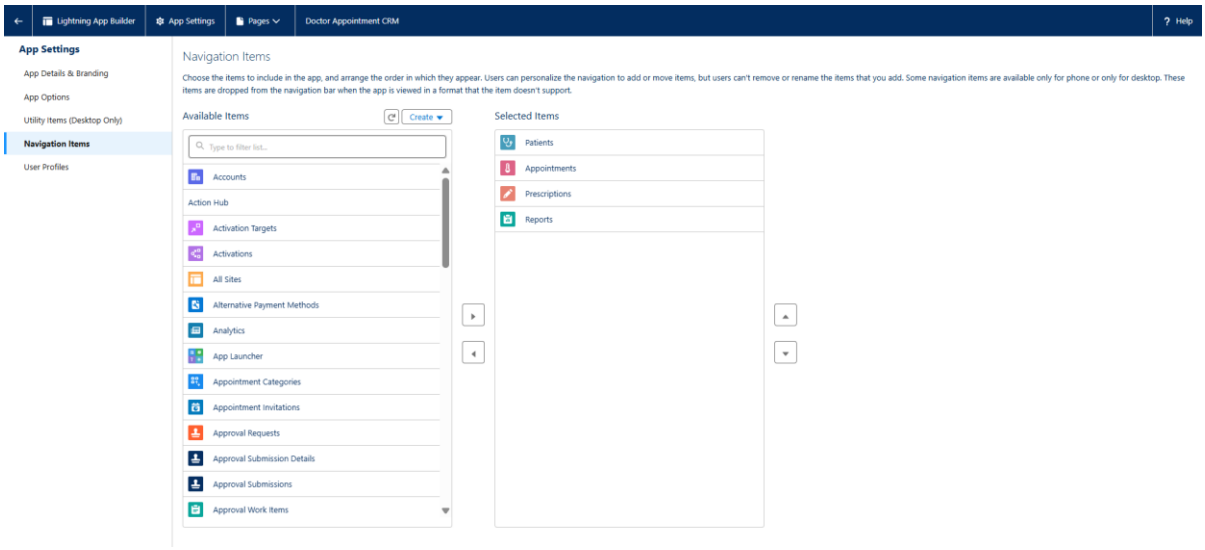
- Added **Highlights Panel** to show key details (Patient Name, Doctor Name, Appointment Date, Status).
- Included **Related Lists** for viewing associated Prescriptions and Appointments.
- Used **Dynamic Components Visibility** to show fields like *Prescription Details* only when the status = “Completed”.



3. Tabs

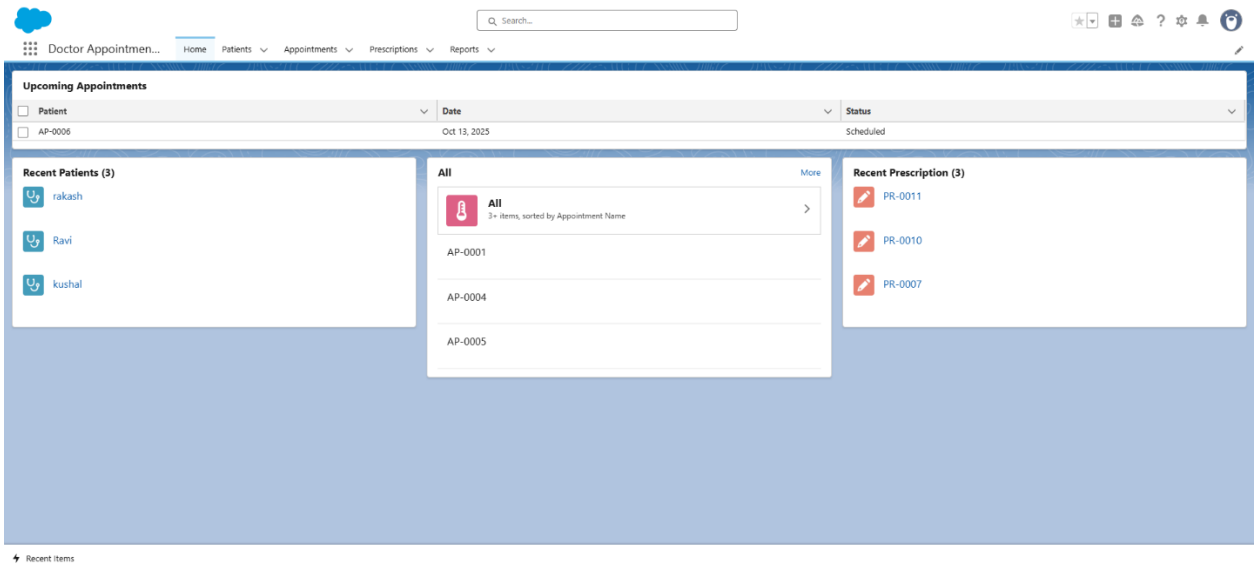
- Created custom tabs for the following custom objects:
  - **Patient__c**
  - **Appointment__c**
  - **Prescription__c**
- These tabs were added to the **Doctor Appointment CRM** app for easy navigation.

- Visibility of tabs configured so Doctors can only see relevant sections.



4. Home Page Layout

- Customized the **Home Page** for the app using Lightning App Builder.
- Added standard components:
    - **Recent Items** – to access recently viewed records.
    - **Dashboard Snapshot** – to show appointment summary and daily schedule.
    - **List View Chart** – displaying count of completed and pending appointments.
- Personalized Home Page made the system dashboard-like for daily use.



5. Utility Bar

- Added a **Utility Bar** to the bottom of the Lightning App for quick access:
    - **“Create New Appointment”** – opens a quick form to add an appointment.
    - **“Help & Support”** – shows FAQ and contact details.
    - **“Notes”** – for doctors to write quick notes.

6. Lightning Web Component (LWC)

A simple **LWC component** was developed to display a doctor’s upcoming appointments dynamically.

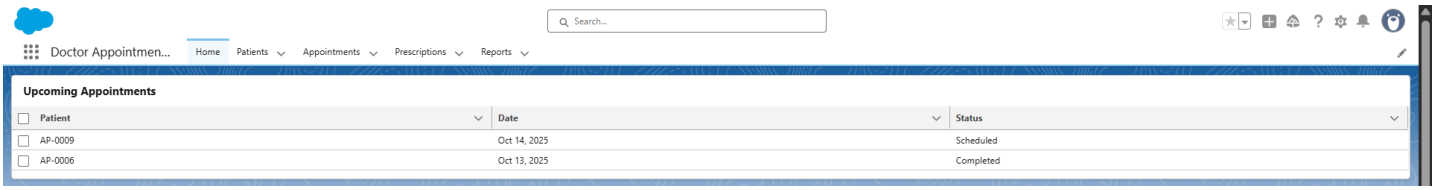
Component Name: doctorUpcomingAppointments

Features:

- Uses **Apex Class (DoctorUtilityV2)** to fetch appointments.
- Displays upcoming appointments in a data table.
- Refreshes automatically when new appointments are created.

Code Snippet (Simplified):

```
// doctorUpcomingAppointments.js
import { LightningElement, wire } from 'lwc';
import getDoctorAppointments from '@salesforce/apex/DoctorUtilityV2.getDoctorAppointments';
export default class DoctorUpcomingAppointments extends LightningElement {
   appointments;
   @wire(getDoctorAppointments, { doctorId: '$userId' })
   wiredAppointments({ data, error }) {
      if (data) this.appointments = data;
   }
}
```



7. Apex with LWC

- The **LWC** used Apex method getDoctorAppointments from DoctorUtilityV2 class.
- The method fetches Appointment records filtered by **Doctor__c** and sorts them by date.

8. Events in LWC

- Implemented simple event handling:
    - When user clicks on an appointment record, a **custom event** navigates to the record page using the Navigation Service.

9. Wire Adapters & Imperative Apex

- Used **@wire adapter** to get data reactively from Apex.
- Tested **imperative Apex calls** to refresh appointment data manually when status changes.

10. Navigation Service

- Implemented Salesforce **NavigationMixin** in LWC to allow redirection to record pages directly.

Example:

```
import { NavigationMixin } from 'lightning/navigation';

this[NavigationMixin.Navigate]({
    type: 'standard__recordPage',
    attributes: {
        recordId: record.Id,
        actionName: 'view'
    }
});
```